

Quadratic Functions — Graphs & Properties

Practice Paper — Digital SAT Style

Difficulty Mix: Easy (Q1–14) | Medium (Q15–30) | Hard (Q31–40) · Total: 40 Questions

DIFFICULTY: EASY

Q1. The graph of $y = x^2 - 6x + 8$ has a vertex at which point?

- A) (3, -1)
 - B) (3, 1)
 - C) (-3, -1)
 - D) (6, 8)
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DIFFICULTY: EASY

Q2. What is the y-intercept of the parabola $y = 2x^2 - 5x + 3$?

- A) 2
 - B) -5
 - C) 3
 - D) 0
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DIFFICULTY: EASY

Q3. The parabola $y = -x^2 + 4x - 3$ opens in which direction?

- A) Upward, because the coefficient of x^2 is negative
 - B) Downward, because the coefficient of x^2 is negative
 - C) Upward, because the constant term is negative
 - D) Downward, because the constant term is negative
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DIFFICULTY: EASY

Q4. Which of the following is an x-intercept of $y = x^2 - 5x + 6$?

- A) $x = 1$
 - B) $x = 2$
 - C) $x = 5$
 - D) $x = 6$
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DIFFICULTY: EASY

Q5. The axis of symmetry of $y = 3x^2 - 12x + 7$ is $x = ?$

- A) $x = -2$
 - B) $x = 2$
 - C) $x = 4$
 - D) $x = 12$
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DIFFICULTY: EASY

Q6. If $f(x) = x^2 - 4$, what are the x-intercepts of the graph of f ?

- A) $x = 4$ only
 - B) $x = -4$ and $x = 4$
 - C) $x = -2$ and $x = 2$
 - D) $x = 2$ only
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DIFFICULTY: EASY

Q7. The vertex of $y = (x - 5)^2 + 3$ is located at:

- A) (5, 3)
 - B) (-5, 3)
 - C) (5, -3)
 - D) (-5, -3)
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DIFFICULTY: EASY

Q8. A parabola has vertex (2, -7) and opens upward. What is the minimum value of the function?

- A) 2
 - B) -7
 - C) 7
 - D) The function has no minimum
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DIFFICULTY: EASY

Q9. Which equation represents a parabola with vertex at the origin?

- A) $y = x^2 + 3$
- B) $y = (x-3)^2$
- C) $y = x^2$

D) $y = x^2 - x$

DIFFICULTY: EASY

Q10. The graph of $y = x^2 - 2x - 3$ crosses the x-axis at $x = -1$ and $x = 3$.

Which factored form matches this?

A) $y = (x+1)(x-3)$

B) $y = (x-1)(x+3)$

C) $y = (x+1)(x+3)$

D) $y = (x-1)(x-3)$

DIFFICULTY: EASY

Q11. If the parabola $y = ax^2 + bx + c$ has $a > 0$, which of the following must be true?

A) The parabola opens downward

B) The parabola has no x-intercepts

C) The parabola has a minimum value

D) The vertex is above the x-axis

DIFFICULTY: EASY

Q12. What is the vertex of the parabola $y = x^2 + 6x + 9$?

A) $(-3, 0)$

B) $(3, 0)$

C) $(-3, 9)$

D) $(0, 9)$

DIFFICULTY: EASY

Q13. How many x-intercepts does the parabola $y = x^2 + 4$ have?

A) 0

B) 1

C) 2

D) 4

DIFFICULTY: EASY

Q14. The function $f(x) = -(x-1)^2 + 9$ achieves its maximum value when $x = ?$

- A) -1
 - B) 0
 - C) 1
 - D) 9
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DIFFICULTY: MEDIUM

Q15. The quadratic $f(x) = x^2 - 8x + k$ has exactly one real root. What is the value of k ?

- A) 4
 - B) 8
 - C) 16
 - D) 64
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DIFFICULTY: MEDIUM

Q16. The graph of $y = 2(x - 3)^2 - 8$ intersects the x -axis at two points.

What are the x -intercepts?

- A) $x = 1$ and $x = 5$
 - B) $x = -1$ and $x = 7$
 - C) $x = 1$ and $x = 7$
 - D) $x = -1$ and $x = 5$
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DIFFICULTY: MEDIUM

Q17. Which of the following quadratics has a vertex in the second quadrant?

- A) $y = (x - 2)^2 + 3$
 - B) $y = (x + 2)^2 + 3$
 - C) $y = (x - 2)^2 - 3$
 - D) $y = (x + 2)^2 - 3$
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DIFFICULTY: MEDIUM

Q18. A ball is thrown upward and its height in feet after t seconds is

$h(t) = -16t^2 + 64t + 5$. What is the maximum height reached by the ball?

- A) 5 feet
- B) 64 feet
- C) 69 feet

D) 128 feet

DIFFICULTY: MEDIUM

Q19. The parabola $y = x^2 - 4x + c$ is tangent to the x-axis (touches but does not cross).

What is the value of c ?

- A) 2
 - B) 4
 - C) 8
 - D) -4
-

DIFFICULTY: MEDIUM

Q20. If the vertex of $y = a(x - 2)^2 + k$ is $(2, -5)$ and the parabola passes through $(4, 3)$,

what is the value of a ?

- A) 1
 - B) 2
 - C) 3
 - D) 4
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DIFFICULTY: MEDIUM

Q21. Which value of k causes the parabola $y = x^2 + kx + 9$ to have exactly one x-intercept?

- A) $k = 3$
 - B) $k = 6$
 - C) $k = -6$
 - D) Both B and C
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DIFFICULTY: MEDIUM

Q22. The graph of $f(x) = -2x^2 + 12x - 10$ has its vertex at $(3, m)$.

What is the value of m ?

- A) 8
 - B) 10
 - C) 12
 - D) 18
-

DIFFICULTY: MEDIUM

Q23. The quadratic equation $x^2 - 6x + 7 = 0$ has two real solutions.

Which best describes the vertex of $y = x^2 - 6x + 7$?

- A) Below the x-axis
 - B) On the x-axis
 - C) Above the x-axis
 - D) The parabola opens downward
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DIFFICULTY: MEDIUM

Q24. A parabola with equation $y = ax^2 + bx + c$ passes through (0, 4), (1, 1), and (2, 4).

What is the axis of symmetry?

- A) $x = 0$
 - B) $x = 1$
 - C) $x = 2$
 - D) $x = 3$
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DIFFICULTY: MEDIUM

Q25. The revenue function for a product is $R(x) = -5x^2 + 200x$, where x is the price in dollars.

At what price is revenue maximised?

- A) \$10
 - B) \$20
 - C) \$40
 - D) \$200
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DIFFICULTY: MEDIUM

Q26. If $f(x) = x^2 - 10x + 21$, what is the range of f ?

- A) $y \geq -4$
 - B) $y \geq 4$
 - C) $y \geq -25$
 - D) All real numbers
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DIFFICULTY: MEDIUM

Q27. The parabola $y = 3x^2 - 6x + 5$ can be rewritten in vertex form as:

A) $y = 3(x-1)^2 + 2$

B) $y = 3(x-1)^2 - 2$

C) $y = 3(x+1)^2 + 2$

D) $y = 3(x-2)^2 + 5$

DIFFICULTY: MEDIUM

Q28. A quadratic function f has zeros at $x = -3$ and $x = 7$ and a leading coefficient of 2.

Which equation represents f ?

A) $f(x) = 2(x+3)(x-7)$

B) $f(x) = 2(x-3)(x+7)$

C) $f(x) = (x+3)(x-7)$

D) $f(x) = 2(x+3)(x+7)$

DIFFICULTY: MEDIUM

Q29. The graph of $y = x^2 - bx + 16$ is tangent to the x -axis. What are the possible values of b ?

A) $b = 4$ only

B) $b = -4$ only

C) $b = 8$ only

D) $b = 8$ or $b = -8$

DIFFICULTY: MEDIUM

Q30. If the parabola $y = x^2 - 4x + k$ lies entirely above the x -axis, which must be true?

A) $k > 2$

B) $k > 4$

C) $k < 4$

D) $k = 4$

DIFFICULTY: HARD

Q31. The function $f(x) = x^2 + px + q$ has vertex at $(-p/2, q - p^2/4)$.

If f has roots at $x = 1$ and $x = 7$, what is the minimum value of f ?

A) -9

B) -8

C) 0

D) 9

DIFFICULTY: HARD

Q32. In the xy -plane, the parabola $y = x^2 + k$ is reflected over the x -axis.

The reflected parabola passes through $(2, -9)$. What is the value of k ?

- A) -9
 - B) -5
 - C) 5
 - D) 9
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DIFFICULTY: HARD

Q33. The quadratic $f(x) = ax^2 + bx + c$ satisfies $f(0) = 5$, $f(1) = 2$, and $f(-1) = 10$.

What is the value of $a + b + c$?

- A) 2
 - B) 5
 - C) 10
 - D) 17
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DIFFICULTY: HARD

Q34. For what values of m does $y = x^2 - 2x + m$ have NO real x -intercepts?

- A) $m < 1$
 - B) $m = 1$
 - C) $m > 1$
 - D) $m > -1$
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DIFFICULTY: HARD

Q35. The parabola $y = 2x^2 - kx + 8$ is tangent to the x -axis.

For which value of k does this occur?

- A) $k = 4$
 - B) $k = 8$
 - C) $k = -8$
 - D) $k = 8$ or $k = -8$
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DIFFICULTY: HARD

Q36. A quadratic function $f(x)$ has vertex $(3, -4)$. The point $(5, 0)$ lies on the graph.

What is the value of $f(1)$?

- A) -4
 - B) -2
 - C) 0
 - D) 4
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DIFFICULTY: HARD

Q37. The sum of two numbers is 20. Their product P is to be maximised.

What is the maximum value of P ?

- A) 19
 - B) 96
 - C) 100
 - D) 400
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DIFFICULTY: HARD

Q38. In the xy -plane, the graphs of $y = x^2 - 3x - 4$ and $y = x + k$ intersect at exactly one point.

What is the value of k ?

- A) -8
 - B) -5
 - C) 5
 - D) 8
-

DIFFICULTY: HARD

Q39. The function $g(x) = -x^2 + 6x - 5$ is defined for $0 \leq x \leq 6$.

What is the range of g on this interval?

- A) $-5 \leq g \leq 4$
 - B) $-5 \leq g \leq 5$
 - C) $0 \leq g \leq 4$
 - D) $0 \leq g \leq 5$
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DIFFICULTY: HARD

Q40. If $f(x) = 3x^2 - 12x + c$ and the minimum value of f is 7, what is the value of c ?

- A)** 7
 - B)** 12
 - C)** 19
 - D)** 25
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